

Assignment 2 – Struct and Buffer

Description:

This assignment is to write a C program that gets the message from the command line argument, puts it in the buffer and outputs the result. Also the assignment requires us to populate personal info, and also put it in the buffer.

Approach:

First, I divided this problem into 2 parts, first one being to populate the personal info structure, and second is to copy the message from the argument. To populate the first name, second name and the message, I took the values from the command line. The make file inputs the first name, second name and the message automatically, so it was convenient to copy it from the CLI. I approached the second part of the problem knowing that the message might be more than 99 characters, so I needed to truncate it. For buffering, I decided to go with nested looping because if I used just one loop, I would either overflow the buffer, or would have to truncate the message, which could lead to data loss. The inner loop ensures that I only copy what fits, commit the block and start over.

Issues and Resolutions:

The first issue I was not familiar with the enum data type, so I had to look it up in order to populate the level data field. Enum contains a set of constants, and helps to assign names to the integers. Another issue I had was that along with the message, I copied the null terminators to the buffer. It gave me a check failed (-1, 97) error. I resolved it by not copying null terminators from the message to the buffer, but filling the buffer with null terminators once the message was complete, and there still was space in the buffer

Analysis:

```
END-OF-ASSIGNMENT
000000: B1 60 5E F1 FF FF 00 00 B6 60 5E F1 FF FF 00 00 | ?^????..?^????..
000010: DD 32 04 37 0A 00 00 00 1F 0C 04 00 46 6F 75 72 | ?2.7.....Four
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E | score and seven
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 | years ago our f
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 | athers brought f
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E | orth on this con
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 | tinent, a new na
000070: 74 69 6F 6E 2C 20 63 6F 6F 63 65 69 76 65 64 00 | tion, conceived.
```

Green - Represents the address of the Firstname pointer

Yellow - Represents the address of the Lastname pointer

Red - Represents student id. 0x370432DD = 0011 0111 0000 0100 0011 0010 1101 1101 = 923022045

Blue - Represents level. 00 00 00 0A = 10 in decimal, which is senior in enum.

Light Blue - represents the knowledge of languages. 0x00040C1F = 0000 0000 0000 0100 0000 1100 0001 1111. Here, each bit represents the language.

Pink - represents the message. It is hexadecimal for each character. For example, 46 is $16 \times 4 + 1 \times 6 = 70$, which is letter F in ASCII table. 6F = $16 \times 6 + 15 = 111$, which is o, and so on.

White represents the null terminator.

Screen shot of compilation:

```
student@student:~/CSC415/Assignment2/csc415-assignment-2-bufferandstruct-vincivv$ make
gcc -c -o Almat_Dias_HW2_main.o Almat_Dias_HW2_main.c -g -rdynamic -I.
gcc -o Almat_Dias_HW2_main Almat_Dias_HW2_main.o assignment2M1.o -g -rdynamic -I.
student@student:~/CSC415/Assignment2/csc415-assignment-2-bufferandstruct-vincivv$
```

Screen shot(s) of the execution of the program:

```
student@student:~/CSC415/Assignment2/csc415-assignment-2-bufferandstruct-vincivv$ make run
./Almat_Dias_HW2_main Dias Almat "Four score and seven years ago our fathers brought forth on thi
s continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men ar
e created equal."
----- CHECK -----
Running the check for Dias Almat
Name check is 0 by 0
Student ID: 923022045, Grade Level: Senior
Languages: 265247
Message:
Four score and seven years ago our fathers brought forth on this continent, a new nation, conceiv
ed

The Check Succeeded (0, 0)

END-OF-ASSIGNMENT
000000: B1 D0 80 F3 FF FF 00 00 B6 D0 80 F3 FF FF 00 00 | ??????..?????..
000010: DD 32 04 37 0A 00 00 00 1F 0C 04 00 46 6F 75 72 | ?2.7.....Four
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E |  score and seven
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 |  years ago our f
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 |  athers brought f
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E |  orth on this con
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 |  tinent, a new na
000070: 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64 00 |  tion, conceived.

student@student:~/CSC415/Assignment2/csc415-assignment-2-bufferandstruct-vincivv$
```